

Math 4102 - Midterm Problem List

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Problem 1: Theorem 30.1

State and prove Egoroff's theorem.

Solution:

30.1 Theorem (Egoroff): Let A be a bounded measurable set in $\mathbb{R}$. Let f_n and f be functions defined on A such that each f_n is measurable and $f_n \rightarrow f$ almost everywhere on A . Then given any $\varepsilon > 0$, there exists a measurable subset $A_\varepsilon \subset A$ such that $m(A \setminus A_\varepsilon) < \varepsilon$ and $f_n \rightarrow f$ uniformly on A_ε .

Proof: Given $\varepsilon > 0$ and positive integers m and n , define the set

$$E_{m,n} = \bigcap_{i=n}^{\infty} \{x \mid |f_i(x) - f(x)| < 1/m\}.$$

Note that $E_{m,n}$ is measurable. If U is the subset on which $f_n \rightarrow f$, then clearly for any m ,

$$U \subset \bigcup_{n=1}^{\infty} E_{m,n} \subset A.$$

Now $m(U) = m(A)$ so

$$m\left(\bigcup_{n=1}^{\infty} E_{m,n}\right) = m(A).$$

But $E_{m,n} \subset E_{m,n+1}$ for all m and n , so

$$\lim_{n \rightarrow \infty} m(A \setminus E_{m,n}) = \lim_{n \rightarrow \infty} [m(A) - m(E_{m,n})] = m(A) - \lim_{n \rightarrow \infty} m(E_{m,n}) = m(A) - m\left(\bigcup_{n=1}^{\infty} E_{m,n}\right) = 0.$$

Thus for each m , there exists an integer n_m such that

$$m(A \setminus E_{m,n_m}) < \varepsilon/2^m.$$

Now let

$$A_\varepsilon = \bigcap_{m=1}^{\infty} E_{m,n_m}.$$

Then A_ε is measurable and

$$m(A \setminus A_\varepsilon) = m\left(A \setminus \bigcap_{m=1}^{\infty} E_{m,n_m}\right) = m\left(\bigcup_{m=1}^{\infty} (A \setminus E_{m,n_m})\right) \leq \sum_{m=1}^{\infty} m(A \setminus E_{m,n_m}) < \sum_{m=1}^{\infty} \varepsilon/2^m = \varepsilon.$$

We now claim that $f_n \rightarrow f$ uniformly on A_ε since given any m , there exists an n_m such that for all $n > n_m$, $|f_n(x) - f(x)| < 1/m$ everywhere on E_{m,n_m} . But $A_\varepsilon \subset E_{m,n_m}$ for every m , so for all $n > n_m$, $|f_n(x) - f(x)| < 1/m$ everywhere on A_ε . But this is exactly uniform convergence. $\square$

Problem 2: Theorem 30.3

State and prove Lusin's theorem.

Solution:

30.3 Theorem (Lusin): If f is a measurable function on a bounded measurable set A , then given any $\varepsilon > 0$, there exists a closed set $C_\varepsilon \subset A$ such that $m(A \setminus C_\varepsilon) < \varepsilon$ and f is continuous on C_ε . (That is, the restriction of f to C_ε is continuous.)

Proof: First, let $f = \sum_{k=1}^n a_k \chi_{E_k}$ be a simple function on A . Given $\varepsilon > 0$, for each k there exists a closed set $C_k \subset E_k$ such that

$$m(E_k \setminus C_k) < \varepsilon/n$$

(see Corollary 13.3). Now $C = \bigcup_{k=1}^n C_k$ is closed, $m(A \setminus C) < \varepsilon$ (Exercise 31.23) and f is continuous on C since C is equal to the disjoint union of closed sets C_k and f is constant on each C_k . (Exercise 31.24).

Now let f be any measurable function. Since $f = f_+ - f_-$, we may assume f is non-negative. Since f is measurable, $f = \lim_{n \rightarrow \infty} f_n$ for a sequence $\{f_n\}$ of simple functions by Theorem 19.4.

By the first part of the proof, given $\varepsilon > 0$, there exists for each n a closed set C_n such that $m(A \setminus C_n) < \varepsilon/2^{n+1}$ and f_n is continuous on C_n .

Let

$$C_0 = \bigcap_{n=1}^{\infty} C_n.$$

Then C_0 is closed and

$$m(A \setminus C_0) = m\left(A \setminus \bigcap_{n=1}^{\infty} C_n\right) = m\left(\bigcup_{n=1}^{\infty} (A \setminus C_n)\right) \leq \sum_{n=1}^{\infty} m(A \setminus C_n) < \varepsilon \sum_{n=1}^{\infty} \frac{1}{2^{n+1}} = \varepsilon/2.$$

We would like to conclude that f is continuous on C_0 , but we cannot since we need something like uniform convergence for this. However, by Egoroff's Theorem, there is a measurable subset $C \subset C_0$ such that $m(C_0 \setminus C) < \varepsilon/4$ [and thus $m(A \setminus C) = m(A \setminus C_0) + m(C_0 \setminus C) < 3\varepsilon/4$], and $f_n \rightarrow f$ uniformly on C .

Now each f_n is continuous on $C \subset C_0$ so that f , being the uniform limit of the f_n , is continuous on C . If C is closed we are through. If not, take closed set $F \subset C$ such that $m(C \setminus F) < \varepsilon/4$. $\square$

Problem 3: Definition 32.3

State the definition of a normed linear space and give both finite- and ∞ -dimensional examples.

Solution:

Definition. A real *normed linear space* is a vector space V (with addition and scalar multiplication denoted by $\bar{u} + \bar{v}$, $a\bar{u}$, respectively) such that for each $\bar{u} \in V$, there is a unique real number $\|\bar{u}\|$ (called the norm or length of $\bar{u}$) satisfying:

1. $\|\bar{u}\| \geq 0$ for all $\bar{u} \in V$;
2. $\|\bar{u}\| = 0$ if and only if $\bar{u} = \bar{0}$;
3. $\|a\bar{u}\| = |a| \|\bar{u}\|$ for all $a \in \mathbb{R}$, $\bar{u} \in V$;
4. $\|\bar{u} + \bar{v}\| \leq \|\bar{u}\| + \|\bar{v}\|$ for all $\bar{u}, \bar{v} \in V$.

Examples.

(Finite-dimensional example) For $(x_1, \dots, x_n) \in \mathbb{R}^n$,

$$\|(x_1, \dots, x_n)\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

(∞ -dimensional example) For $f \in C([0, 1])$,

$$\|f\| = \int_0^1 |f(x)| dx.$$

Problem 4: Example 32.15

State the definition of a real Hilbert space and give both finite- and ∞ -dimensional examples.

Solution:

A real *Hilbert space* is a real inner product space V that is complete with respect to the norm $\|\bar{u}\| = \sqrt{\bar{u} \cdot \bar{u}}$ induced by the inner product.

Examples.

(Finite-dimensional example) $\mathbb{R}^n$ with the standard inner product

$$\bar{u} \cdot \bar{v} = \sum_{i=1}^n u_i v_i$$

is a Hilbert space. Completeness follows from the completeness of $\mathbb{R}$.

(∞ -dimensional example) $L^2[a, b]$ with the inner product

$$f \cdot g = \int_{[a, b]} fg \, dm$$

is a Hilbert space. Completeness is given by the Riesz–Fischer theorem (Theorem 33.11).

Problem 5

Determine whether $\{\cos kx : k = 0, 1, 2, \dots\}$ is an orthogonal set in $L^2[0, \pi]$.

Solution:

For $j, k = 0, 1, 2, \dots$, consider

$$\int_0^\pi \cos jx \cos kx \, dx.$$

If $j \neq k$, then by the product-to-sum identity,

$$\cos jx \cos kx = \frac{1}{2} [\cos((j-k)x) + \cos((j+k)x)].$$

Therefore,

$$\begin{aligned} \int_0^\pi \cos jx \cos kx \, dx &= \frac{1}{2} \int_0^\pi \cos((j-k)x) \, dx + \frac{1}{2} \int_0^\pi \cos((j+k)x) \, dx \\ &= \frac{1}{2} \left[\frac{\sin((j-k)x)}{j-k} \right]_0^\pi + \frac{1}{2} \left[\frac{\sin((j+k)x)}{j+k} \right]_0^\pi \\ &= 0. \end{aligned}$$

Also, none of the functions $\cos kx$ is the zero function in $L^2[0, \pi]$. Hence

$$\{\cos kx : k = 0, 1, 2, \dots\}$$

is an orthogonal set in $L^2[0, \pi]$.

Problem 6: Theorem 32.13

State and prove Cauchy's inequality

Solution:

32.13 Cauchy Inequality: If V is an inner product space with inner product $\bar{u} \cdot \bar{v}$, then for any $\bar{u}, \bar{v} \in V$,

$$|\bar{u} \cdot \bar{v}| \leq \sqrt{\bar{u} \cdot \bar{u}} \sqrt{\bar{v} \cdot \bar{v}}.$$

Proof: If $\bar{u} = \bar{0}$ or $\bar{v} = \bar{0}$, then equality holds trivially. Otherwise, consider the vectors

$$\bar{u}' = \frac{\bar{u}}{\sqrt{\bar{u} \cdot \bar{u}}} \quad \text{and} \quad \bar{v}' = \frac{\bar{v}}{\sqrt{\bar{v} \cdot \bar{v}}}.$$

It is easy to verify that $\bar{u}' \cdot \bar{u}' = \bar{v}' \cdot \bar{v}' = 1$, so that by the lemma,

$$|\bar{u}' \cdot \bar{v}'| \leq 1.$$

But

$$|\bar{u}' \cdot \bar{v}'| = \frac{|\bar{u} \cdot \bar{v}|}{\sqrt{\bar{u} \cdot \bar{u}} \sqrt{\bar{v} \cdot \bar{v}}} \leq 1.$$

□

Problem 7: Theorem 33.9

State and prove Minkowsky's inequality in $\mathcal{L}^2$.

Solution:

Theorem (Minkowski Inequality in L^2). For $f, g \in L^2(A)$,

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2,$$

i.e.,

$$\sqrt{\int_A (f + g)^2 dm} \leq \sqrt{\int_A f^2 dm} + \sqrt{\int_A g^2 dm}.$$

Proof: We expand $\|f + g\|_2^2$:

$$\|f + g\|_2^2 = \int_A (f + g)^2 dm = \int_A f^2 dm + 2 \int_A fg dm + \int_A g^2 dm.$$

By the Cauchy inequality (Theorem 33.7),

$$\int_A fg dm \leq \left| \int_A fg dm \right| \leq \|f\|_2 \|g\|_2.$$

Therefore,

$$\|f + g\|_2^2 \leq \|f\|_2^2 + 2\|f\|_2 \|g\|_2 + \|g\|_2^2 = (\|f\|_2 + \|g\|_2)^2.$$

Taking square roots of both nonnegative sides gives

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2.$$

□

Problem 8: Page 120

Define a trigonometric polynomial and a trigonometric series

Solution:

A **trigonometric polynomial** is a finite sum of the form

$$\sum_{k=0}^n (a_k \cos kx + b_k \sin kx),$$

where $a_k, b_k \in \mathbb{R}$.

A **trigonometric series** is an infinite series of the form

$$\sum_{k=0}^{\infty} (a_k \cos kx + b_k \sin kx),$$

where $a_k, b_k \in \mathbb{R}$.

Problem 9: Theorem 36.3

State and prove Bessel's inequality for an orthonormal set in $L^2[-\pi, \pi]$.

Solution:

Let $\{u_k\}_{k=0}^{\infty}$ be an orthonormal set in $L^2[-\pi, \pi]$, and let $f \in L^2[-\pi, \pi]$. Define

$$c_k = f \cdot u_k = \int_{[-\pi, \pi]} f u_k \, dm.$$

Then Bessel's inequality states that

$$\sum_{k=0}^{\infty} c_k^2 \leq \int_{[-\pi, \pi]} f^2 \, dm.$$

Proof. By Corollary 36.2, for every n ,

$$\int_{[-\pi, \pi]} f^2 \, dm \geq \sum_{k=0}^n c_k^2.$$

Since each $c_k^2 \geq 0$, the partial sums

$$\sum_{k=0}^n c_k^2$$

are increasing and bounded above by $\int_{[-\pi, \pi]} f^2 \, dm$. Therefore the infinite series converges and satisfies

$$\sum_{k=0}^{\infty} c_k^2 \leq \int_{[-\pi, \pi]} f^2 \, dm.$$

Problem 10: Theorem 36.4

State and prove the Riemann-Lebesgue lemma for the Fourier series of $f \in L^2[-\pi, \pi]$.

Solution:

Let $\{u_k\}_{k=0}^\infty$ be an orthonormal set in $L^2[-\pi, \pi]$, and let $f \in L^2[-\pi, \pi]$. Define

$$c_k = f \cdot u_k = \int_{[-\pi, \pi]} f u_k \, dm.$$

The Riemann-Lebesgue lemma states that

$$\lim_{k \rightarrow \infty} c_k = 0.$$

Proof. By Bessel's inequality,

$$\sum_{k=0}^{\infty} c_k^2 \leq \int_{[-\pi, \pi]} f^2 \, dm.$$

Thus $\sum_{k=0}^{\infty} c_k^2$ is a convergent series of nonnegative terms. Therefore,

$$\lim_{k \rightarrow \infty} c_k^2 = 0.$$

Hence,

$$\lim_{k \rightarrow \infty} c_k = 0.$$

Problem 11: Lemma 37.4

Give two formulas for the Dirichlet kernel $D_n(t)$.

Solution:

The **Dirichlet kernel** $D_n(t)$ is

$$D_n(t) = \frac{1}{2} + \sum_{k=1}^n \cos(kt).$$

Equivalently, using Lemma 37.4,

$$D_n(t) = \begin{cases} 0, & t = 2k\pi, \ k \in \mathbb{Z}, \\ \frac{\sin\left((n + \frac{1}{2})t\right)}{2 \sin\left(\frac{t}{2}\right)}, & \text{otherwise.} \end{cases}$$

Problem 12: Lemma 37.5

Express the partial sum $s_n(x)$ of the Fourier series for a function $f \in L^2[-\pi, \pi]$ using the Lebesgue integral and the Dirichlet kernel.

Solution:

By Lemma 37.5, if $f \in L^2[-\pi, \pi]$, then the Fourier partial sum $s_n(x)$ can be written as

$$s_n(x) = \frac{1}{\pi} \int_0^\pi [f(x+t) + f(x-t)] D_n(t) dt,$$

where

$$D_n(t) = \begin{cases} 0, & t = 2k\pi, \ k \in \mathbb{Z}, \\ \frac{\sin((n + \frac{1}{2})t)}{2 \sin(\frac{t}{2})}, & \text{otherwise.} \end{cases}$$

Here f is understood to be extended periodically with period 2π .

Problem 13: Corollary 37.6

Give a formula for the Fejér kernel $K_n(t)$ and relate it to the Dirichlet kernel.

Solution:

The **Fejér kernel** $K_n(t)$ is

$$K_n(t) = \begin{cases} 0, & t = 2m\pi, \ m \in \mathbb{Z}, \\ \frac{\sin^2\left(\frac{nt}{2}\right)}{2n \sin^2\left(\frac{t}{2}\right)}, & \text{otherwise.} \end{cases}$$

Its relation with the Dirichlet kernel is

$$K_n(t) = \frac{1}{n} \sum_{k=0}^{n-1} D_k(t).$$

Indeed, using the definition of $D_k(t)$,

$$\frac{1}{n} \sum_{k=0}^{n-1} D_k(t) = \frac{1}{2n \sin(t/2)} \sum_{k=0}^{n-1} \sin\left(\left(k + \frac{1}{2}\right)t\right).$$

By Lemma 37.4,

$$\sum_{k=0}^{n-1} \sin\left(\left(k + \frac{1}{2}\right)t\right) = \sum_{j=1}^n \sin\left((2j-1)\frac{t}{2}\right) = \frac{\sin^2(nt/2)}{\sin(t/2)}.$$

Therefore,

$$\frac{1}{n} \sum_{k=0}^{n-1} D_k(t) = \frac{\sin^2(nt/2)}{2n \sin^2(t/2)} = K_n(t).$$

Problem 14: Definition 37.1

Define the Cesàro $(C, 1)$ -sum of a series $\sum_{k=1}^{\infty} a_k$. Give an example of a divergent series with a convergent $(C, 1)$ -sum.

Solution:

A series $\sum_{k=1}^{\infty} a_k$ is said to have $(C, 1)$ -**sum** (or Cesàro sum) s if

$$\lim_{n \rightarrow \infty} \sigma_n = s,$$

where

$$\sigma_n = \frac{s_1 + s_2 + \cdots + s_n}{n} \quad \text{and} \quad s_n = \sum_{k=1}^n a_k.$$

The sequence $\{\sigma_n\}$ is called the sequence of arithmetic means.

An example is the series

$$1 - 1 + 1 - 1 + \cdots,$$

which diverges, but has $(C, 1)$ -sum $\frac{1}{2}$, since

$$\sigma_n = \begin{cases} \frac{1}{2}, & \text{if } n \text{ is even,} \\ \frac{n+1}{2n}, & \text{if } n \text{ is odd,} \end{cases}$$

and hence $\sigma_n \rightarrow \frac{1}{2}$ as $n \rightarrow \infty$.

Problem 15: Corollary 37.6

Express the $(C, 1)$ -sum $\sigma_n(x)$ of the Fourier series for a function $f \in L[-\pi, \pi]$ using the Lebesgue integral and the Fejér kernel.

Solution:

Let

$$\sigma_n(x) = \frac{s_0(x) + s_1(x) + \cdots + s_{n-1}(x)}{n},$$

where $s_n(x)$ is the n th Fourier partial sum. By Corollary 37.6,

$$\sigma_n(x) = \frac{1}{\pi} \int_0^\pi [f(x+t) + f(x-t)] K_n(t) dt,$$

where $K_n(t)$ is the Fejér kernel of order n , given by

$$K_n(t) = \begin{cases} 0, & t = 2m\pi, \ m \in \mathbb{Z}, \\ \frac{\sin^2\left(\frac{nt}{2}\right)}{2n \sin^2\left(\frac{t}{2}\right)}, & \text{otherwise.} \end{cases}$$

Problem 16: Theorem 37.8

Prove that if $f \in L^2[-\pi, \pi]$ and f is continuous, then $\{\sigma_n\}$ converges uniformly to f .

Solution:

Since f is continuous on $[-\pi, \pi]$, it is bounded and uniformly continuous. Thus there exists a number M such that

$$|f(x)| < M$$

for all $x \in [-\pi, \pi]$. By uniform continuity, given $\varepsilon > 0$, there exists a $\delta > 0$ such that $|x - y| < \delta$ in $[-\pi, \pi]$ implies

$$|f(x) - f(y)| < \frac{\varepsilon}{2}.$$

We will show that for some N , $n > N$ implies

$$|f(x) - \sigma_n(x)| < \varepsilon$$

for every $x \in [-\pi, \pi]$.

Now

$$\begin{aligned} |\sigma_n(x) - f(x)| &\stackrel{37.6}{=} \left| \frac{2}{\pi} \int_0^\pi \frac{f(x+t) + f(x-t)}{2} K_n(t) dt - f(x) \right| \\ &\stackrel{37.7}{=} \left| \frac{2}{\pi} \int_0^\pi \frac{f(x+t) + f(x-t)}{2} K_n(t) dt - f(x) \frac{2}{\pi} \int_0^\pi K_n(t) dt \right| \\ &\leq \left| \frac{2}{\pi} \int_0^\delta \frac{f(x+t) + f(x-t) - 2f(x)}{2} K_n(t) dt \right| \\ &\quad + \left| \frac{2}{\pi} \int_\delta^\pi \frac{f(x+t) + f(x-t) - 2f(x)}{2} K_n(t) dt \right| \\ &\leq \frac{\varepsilon}{2} \frac{2}{\pi} \int_0^\delta K_n(t) dt \\ &\quad + \frac{2}{4n\pi \sin^2(\delta/2)} \int_\delta^\pi [|f(x+t)| + |f(x-t)| + 2|f(x)|] dt \\ &< \frac{\varepsilon}{2} + \frac{1}{2n\pi \sin^2(\delta/2)} \int_\delta^\pi 4M dt. \end{aligned}$$

The second term is clearly less than $\varepsilon/2$ for n large enough, so the result follows.

Problem 17: Theorem 40.2

Prove that if $f \in L^2[-\pi, \pi]$ and f is differentiable at x , then $\{s_n(x)\}$ converges to $f(x)$.

Solution:

Let $x_0 \in [-\pi, \pi]$ and suppose f is differentiable at x_0 . From Lemma 37.5, the n th Fourier partial sum is

$$s_n(x_0) = \frac{1}{\pi} \int_0^\pi [f(x_0 + t) + f(x_0 - t)] D_n(t) dt.$$

Also, for the particular case $f(x) \equiv 1$, we obtain

$$1 = \frac{2}{\pi} \int_0^\pi D_n(t) dt \quad \text{for } n = 1, 2, 3, \dots$$

Thus, for our general f , differentiable at x_0 ,

$$s_n(x_0) - f(x_0) = \frac{1}{\pi} \int_0^\pi [f(x_0 + t) + f(x_0 - t) - 2f(x_0)] D_n(t) dt.$$

Now define

$$h(t) = \frac{f(x_0 + t) + f(x_0 - t) - 2f(x_0)}{2 \sin \frac{t}{2}}.$$

Then $h(t)$ can be defined so as to be continuous at $t = 0$. Therefore, $h \in L^2[0, \pi]$. Thus,

$$s_n(x_0) - f(x_0) = \frac{1}{\pi} \int_0^\pi h(t) \sin \left(\left(n + \frac{1}{2} \right) t \right) dt.$$

Thus by the Riemann–Lebesgue Lemma (Corollary 36.4),

$$s_n(x_0) - f(x_0) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Problem 18: Theorem 41.2

State and prove the Riemann localization lemma.

Solution:

Let $f \in L[-\pi, \pi]$ and $x \in [-\pi, \pi]$, and let δ be any positive real number. Then

$$\lim_{n \rightarrow \infty} \int_{\delta}^{\pi} [f(x+t) + f(x-t)] D_n(t) dt = 0.$$

Proof. Consider

$$g(t) = \begin{cases} 0, & \text{on } [0, \delta), \\ \frac{f(x+t) + f(x-t)}{2 \sin(t/2)}, & \text{on } [\delta, \pi]. \end{cases}$$

Clearly $g \in L[-\pi, \pi]$ since

$$2 \sin(t/2) \geq 2 \sin(\delta/2) > 0 \quad \text{on } [\delta, \pi].$$

Thus

$$\int_{\delta}^{\pi} [f(x+t) + f(x-t)] D_n(t) dt = \int_0^{\pi} g(t) \sin \left(n + \frac{1}{2} \right) t dt.$$

Therefore,

$$\lim_{n \rightarrow \infty} \int_{\delta}^{\pi} [f(x+t) + f(x-t)] D_n(t) dt = 0$$

by the Riemann–Lebesgue Lemma and Exercise 42.6.

Problem 19: Exercise 19

Compute the complex exponential Fourier series of

$$f(x) = \sin x + 2 \cos 3x.$$

Solution:

We write the complex exponential Fourier series in the form

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

Using Euler's formulas,

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}, \quad \cos 3x = \frac{e^{3ix} + e^{-3ix}}{2}.$$

Therefore,

$$\begin{aligned} f(x) &= \sin x + 2 \cos 3x \\ &= \frac{e^{ix} - e^{-ix}}{2i} + 2 \left(\frac{e^{3ix} + e^{-3ix}}{2} \right) \\ &= -\frac{i}{2} e^{ix} + \frac{i}{2} e^{-ix} + e^{3ix} + e^{-3ix}. \end{aligned}$$

Thus the only nonzero complex Fourier coefficients are

$$c_1 = -\frac{i}{2}, \quad c_{-1} = \frac{i}{2}, \quad c_3 = 1, \quad c_{-3} = 1.$$

All other coefficients are zero. Hence

$$f(x) = \frac{i}{2} e^{-ix} - \frac{i}{2} e^{ix} + e^{-3ix} + e^{3ix}.$$